Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

[AIME-2000-I-Q03] | Part A (Q1-Q5)

Answer: 667

1. The binomial coefficients give $\binom{2000}{2}a^2b^{1998} = \binom{2000}{3}a^3b^{1997}$. Cancelling positive factors gives $b/a = 1998/3 = 666$.

2. Coprimality forces $(a, b) = (1, 666)$, so the sum is 667.

Source / 来源

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

[AIME-2002-I-Q07] | Part B (Q6-Q10)

Answer: 428

1. Let $T = 10^{2002}$. Taylor's formula for $(T+1)^{10/7}$ gives $T^{10/7} + (10/7)T^{3/7} + R$, with $0 < R < (15/49)T^{-4/7}$. Thus the first term is the integer 10^{2860} and the second is $10^{859}/7$, while the positive remainder is below 10^{-1000}.

2. Since $859 \equiv 1 \pmod 6$ and 10 has order 6 modulo 7, the second term has fractional part $3/7 = 0.428571\ldots$.

3. The distance from $3/7$ to the next three-decimal boundary 0.429 exceeds 0.0004, vastly larger than R. Therefore the requested digits are 428.

Source / 来源